

9.



Figure 9.1

Figure 9.1 shows a microwave oven. Mary wants to estimate the useful output power of the oven. She is provided with the apparatus and material shown in Figure 9.2.

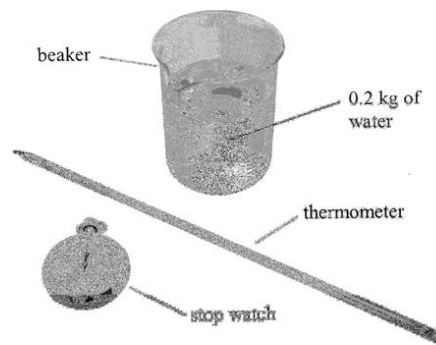


Figure 9.2

- (a) Describe how Mary should conduct the experiment. Specify all measurements Mary has to take. State **EITHER** one precaution taken **OR** one assumption made when conducting this experiment. Write down an equation for calculating the useful output power. (5 marks)

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- (b) The value obtained by Mary is found to be smaller than the rated power of the oven. Suggest one possible reason to account for this difference. (1 mark)

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- (c) Explain whether increasing the mass of water used in the experiment would improve the accuracy of the experiment. (1 mark)

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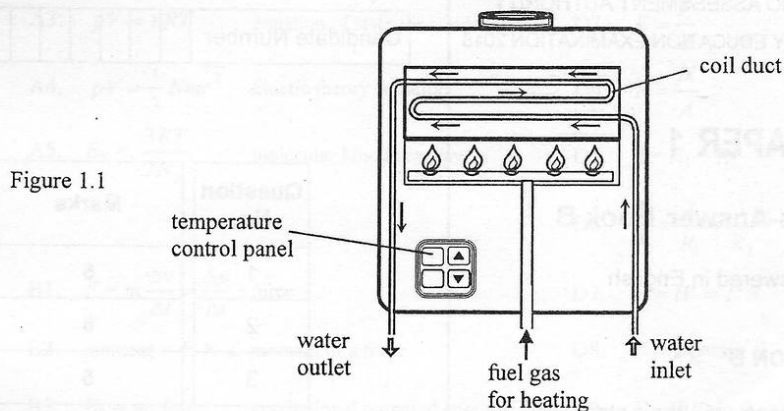
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Section B: Answer **ALL** questions. Parts marked with * involve knowledge of the extension component. Write your answers in the spaces provided.

1. Figure 1.1 shows a domestic water heater. Tap water entering the heater is heated when it passes through the coil duct and exits as hot water at a certain temperature.



On a certain day in winter, the temperature of tap water is 15°C . When the heater is in use, it delivers 6 kg of hot water at a temperature of 50°C in 1 minute . Assume that there is no heat exchange between the heater and the surroundings. Given: specific heat capacity of water = $4200\text{ J kg}^{-1}^\circ\text{C}^{-1}$

- (a) Estimate the power supplied to the tap water by the heater. (3 marks)

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- (b) Assuming the power estimated in (a) remains unchanged, determine the flow rate of tap water into the heater, in kg per minute , such that hot water at a temperature of 40°C is delivered by the heater. (2 marks)

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Book 1 section 2.1: DSE long questions

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